

December

SPECIAL EVENTS

PHASES OF THE MOON

	Full Moon	New Moon
2007	December 24	December 9
2008	December 12	December 27
2009	December 2, 31	December 16
2010	December 21	December 5
2011	December 10	December 24
2012	December 28	December 13

ECLIPSES

2009: December 31 A partial eclipse of the Moon is visible from Australia, Asia, Africa, and Europe.

2010: December 21 A total eclipse of the Moon is visible from North and South America and the Pacific Ocean.

2011: December 10 A total eclipse of the Moon is visible from the Pacific Ocean, Australasia, and Asia.

THE PLANETS

2007: December 24
Mars is at opposition, magnitude -1.6 . At midnight, it is visible to the south from northern latitudes and to the north from southern latitudes.

2008: December 1
Venus and Jupiter appear four Moon-widths apart in the western evening sky, next to a crescent Moon.

2009: December 18
Mercury is at greatest evening elongation, magnitude -0.3 .

2010: December 1
Mercury is at greatest evening elongation, magnitude -0.4 .

2011: December 23
Mercury is at greatest morning elongation, magnitude -0.3 .

2012: December 3
Jupiter is at opposition, magnitude -2.8 . At midnight, it is visible to the south from northern latitudes and to the north from southern latitudes.

2012: December 4
Mercury is at greatest morning elongation, magnitude -0.3 .

Opposition and elongation are explained on [page 14](#)

Taurus is ideally placed for everyone to observe its fine star clusters. This month also offers the first opportunity for a good look at Orion as it moves higher in the sky. The Sun is at its farthest point south of the celestial equator on December 21–22. Northern-latitude nights are the longest of the year, while southern ones are the shortest.

NORTHERN LATITUDES

Winter stars occupy half the view to the south: Taurus leads the way and is high and almost due south; Orion is in the southeast; and Gemini in the eastern sky. Two water constellations, Pisces, the fish, and Cetus, the sea monster, move toward the western horizon. Two bright stars stand out above and below the figure of Orion: yellow Capella, in Auriga and sixth brightest, shines overhead; brilliant-white Sirius, the brightest of all, is near the southeast horizon.

Taurus is easy to see due to its distinctive shape. Its stars depict a bull's head and shoulders; the bright red giant Aldebaran is its eye. Aldebaran appears to be part of the Hyades open cluster in the bull's face but is only half its distance away. The Hyades are easily visible to the naked eye; binoculars reveal many more stars. A second cluster, the Pleiades (M45), marks

the start of the bull's back. About six stars are seen by eye; dozens through binoculars. A telescope shows the Crab Nebula (M1) next to the tip of a horn. It is the remains of a star that exploded in 1054.

To the north, both Ursa Major and Ursa Minor are below Polaris; Andromeda and Perseus remain overhead. A naked-eye knot of light in Perseus's hand is the Double Cluster. The Geminids, the second-best meteor shower, reaches its peak on the 13th, when up to 100 meteors an hour are seen.

Double Cluster

The Double Cluster is a pair of open star clusters within Perseus. NGC 884 is to the left, and NGC 869 to the right. Each contains hundreds of stars. The naked eye sees them as a brighter patch in the path of the Milky Way.

